# Appendices

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This appendix contains the supplementary materials related to the report titled "Analysis of the Relationship Between Customer Demographics, Social Media Usage, and Marketing Campaign Performance."
Tables, figures, and additional calculations referenced throughout the main text are included here to support and illustrate the findings and conclusions.
The numbering of tables and figures corresponds directly to the section numbers used in the main body of the report.

## 2.2 Demographic and Behavioral Patterns Across Social Media Platforms

This section presents demographic and behavioral patterns across different age groups, genders, and social media usage behaviours. The findings are illustrated through a series of tables and figures for clarity.

**Table: 2.2.1** Gender Distribution Across Age Groups

| age_group | gender | count | percentage |
|---|---|---|---|
| 18-25 | Total | 878 | 100 |
| 18-25 | Female | 274 | 31.21 |
| 18-25 | Male | 293 | 33.37 |
| 18-25 | Other | 311 | 35.42 |
| 26-35 | Total | 1031 | 100 |
| 26-35 | Female | 340 | 32.98 |
| 26-35 | Male | 360 | 34.92 |
| 26-35 | Other | 331 | 32.1 |
| 36-45 | Total | 1046 | 100 |
| 36-45 | Female | 353 | 33.75 |
| 36-45 | Male | 355 | 33.94 |
| 36-45 | Other | 338 | 32.31 |
| 46-55 | Total | 1003 | 100 |
| 46-55 | Female | 343 | 34.2 |
| 46-55 | Male | 334 | 33.3 |
| 46-55 | Other | 326 | 32.5 |
| 56-65 | Total | 1042 | 100 |
| 56-65 | Female | 328 | 31.48 |
| 56-65 | Male | 375 | 35.99 |
| 56-65 | Other | 339 | 32.53 |

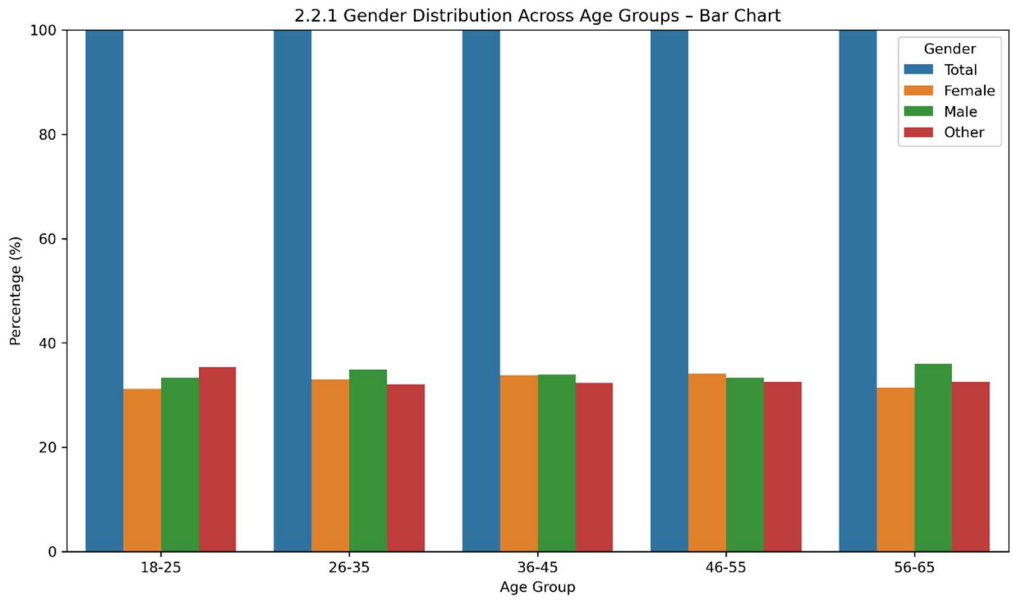


**Figure 2.2.1** Gender Distribution Across Age Groups – Bar Chart

**Table: 2.2.1** Income Level Distribution Across Age Groups

| age_group | income_level | count | percentage |
|---|---|---|---|
| 18-25 | Total | 878 | 100 |
| 18-25 | Low | 292 | 33.26 |
| 18-25 | Medium | 291 | 33.14 |
| 18-25 | High | 295 | 33.6 |
| 26-35 | Total | 1031 | 100 |
| 26-35 | Low | 350 | 33.95 |
| 26-35 | Medium | 327 | 31.72 |
| 26-35 | High | 354 | 34.34 |
| 36-45 | Total | 1046 | 100 |
| 36-45 | Low | 375 | 35.85 |
| 36-45 | Medium | 339 | 32.41 |
| 36-45 | High | 332 | 31.74 |
| 46-55 | Total | 1003 | 100 |
| 46-55 | Low | 316 | 31.51 |
| 46-55 | Medium | 329 | 32.8 |
| 46-55 | High | 358 | 35.69 |
| 56-65 | Total | 1042 | 100 |
| 56-65 | Low | 359 | 34.45 |
| 56-65 | Medium | 371 | 35.6 |
| 56-65 | High | 312 | 29.94 |

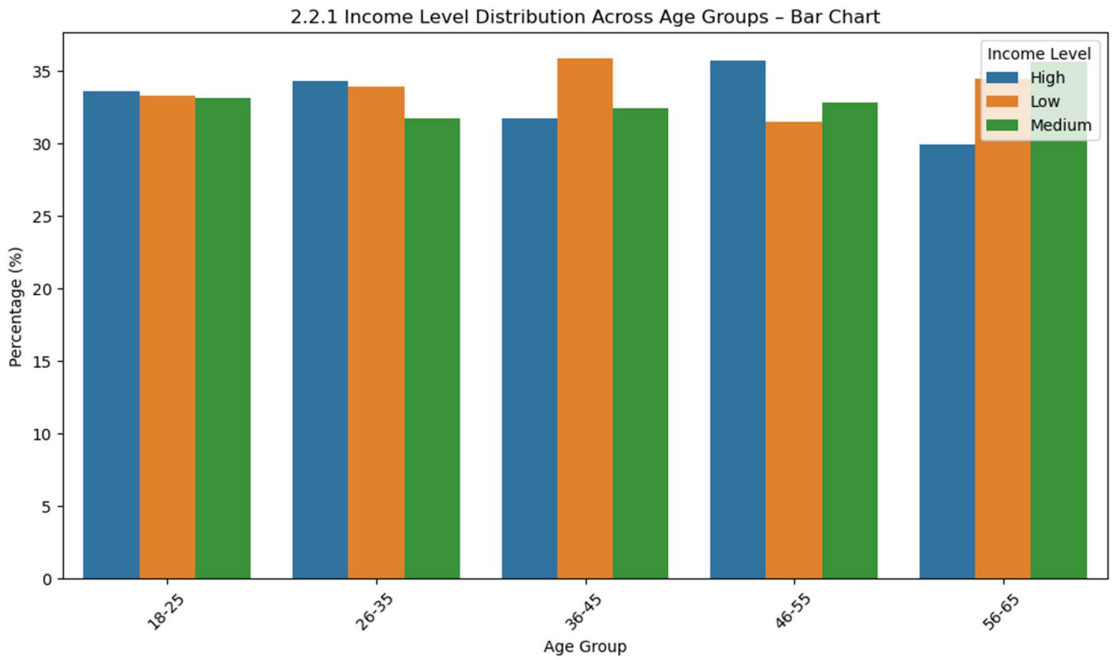


**Figure 2.2.1** Income Level Distribution Across Age Groups – Bar Chart

**Table: 2.2.1** Social Media Platform Usage Across Education Levels

| education level | platform | count | percentage |
|---|---|---|---|
| **Bachelor's** | Facebook | 307 | 24.84 |
| **Bachelor's** | Instagram | 328 | 26.54 |
| **Bachelor's** | LinkedIn | 284 | 22.98 |
| **Bachelor's** | Twitter | 317 | 25.65 |
| **High School** | Facebook | 325 | 25.45 |
| **High School** | Instagram | 335 | 26.23 |
| **High School** | LinkedIn | 301 | 23.57 |
| **High School** | Twitter | 316 | 24.75 |
| **Master's** | Facebook | 307 | 24.46 |
| **Master's** | Instagram | 325 | 25.9 |
| **Master's** | LinkedIn | 312 | 24.86 |
| **Master's** | Twitter | 311 | 24.78 |
| **PhD** | Facebook | 298 | 24.19 |
| **PhD** | Instagram | 313 | 25.41 |
| **PhD** | LinkedIn | 308 | 25 |
| **PhD** | Twitter | 313 | 25.41 |

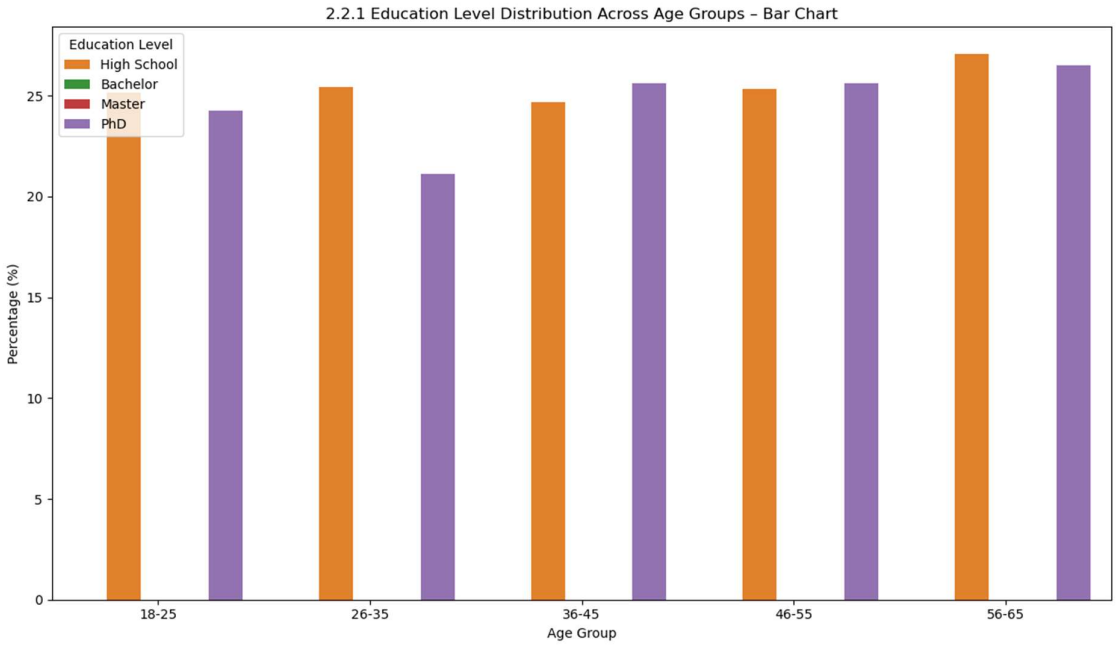


**Figure 2.2.1** Social Media Platform Usage Distribution Across Education Levels – Bar Chart

**Table: 2.2.1** Regional Distribution Across Age Groups

| age_group | location | count | percentage |
|---|---|---|---|
| 18-25 | Canada | 167 | 19.02 |
| 18-25 | Germany | 177 | 20.16 |
| 18-25 | India | 199 | 22.67 |
| 18-25 | Total | 878 | 100 |
| 18-25 | UK | 175 | 19.93 |
| 18-25 | USA | 160 | 18.22 |
| 26-35 | Canada | 216 | 20.95 |
| 26-35 | Germany | 205 | 19.88 |
| 26-35 | India | 206 | 19.98 |
| 26-35 | Total | 1031 | 100 |
| 26-35 | UK | 192 | 18.62 |
| 26-35 | USA | 212 | 20.56 |
| 36-45 | Canada | 220 | 21.03 |
| 36-45 | Germany | 206 | 19.69 |
| 36-45 | India | 217 | 20.75 |
| 36-45 | Total | 1046 | 100 |
| 36-45 | UK | 215 | 20.55 |
| 36-45 | USA | 188 | 17.97 |
| 46-55 | Canada | 195 | 19.44 |
| 46-55 | Germany | 209 | 20.84 |
| 46-55 | India | 190 | 18.94 |
| 46-55 | Total | 1003 | 100 |
| 46-55 | UK | 200 | 19.94 |
| 46-55 | USA | 209 | 20.84 |
| 56-65 | Canada | 213 | 20.44 |
| 56-65 | Germany | 204 | 19.58 |
| 56-65 | India | 227 | 21.79 |
| 56-65 | Total | 1042 | 100 |
| 56-65 | UK | 220 | 21.11 |
| 56-65 | USA | 178 | 17.08 |

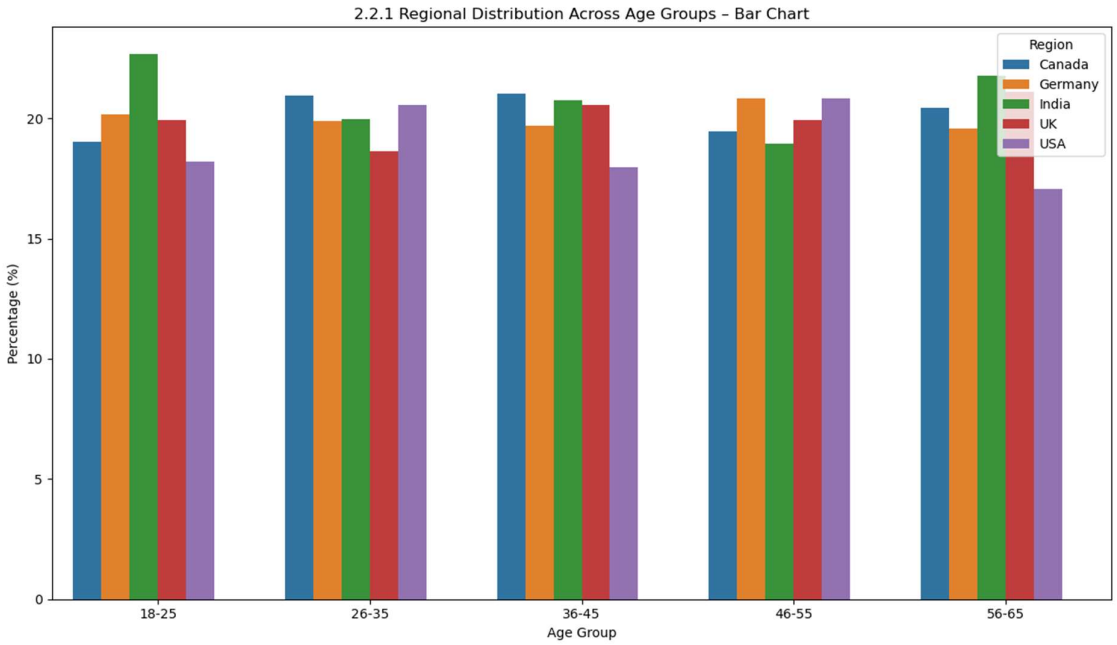


**Figure 2.2.1** Regional Distribution Across Age Groups – Bar Chart

**Table 2.2.1** - Social Media Platform Usage by Region (Counts and Percentages)

| location | **Facebook count** | **Facebook percent** | **Instagram count** | **Instagram percent** | **Linkedln count** | **Linkedln percent** | **Twitter count** | **Twitter percent** |
|---|---|---|---|---|---|---|---|---|
| Canada | 246 | 24.33 | 257 | 25.42 | 265 | 26.21 | 243 | 24.04 |
| Germany | 253 | 25.27 | 265 | 26.47 | 218 | 21.78 | 265 | 26.47 |
| India | 241 | 23.2 | 268 | 25.79 | 260 | 25.02 | 270 | 25.99 |
| UK | 249 | 24.85 | 265 | 26.45 | 241 | 24.05 | 247 | 24.65 |
| USA | 248 | 26.19 | 246 | 25.98 | 221 | 23.34 | 232 | 24.5 |

**Table: 2.2.2** Monthly Customer Behavior Analysis by Age Group

| Age Group | Monthly Average Time Spent Online (hours) | Monthly Average Purchase Frequency | Average Order Value per Purchase (USD) |
|---|---|---|---|
| **18-25** | 104.0862415 | 15.09567198 | 259.947836 |
| **26-35** | 99.99794374 | 14.65955383 | 239.2533172 |
| **36-45** | 106.0045889 | 14.37571702 | 252.9492161 |
| **46-55** | 100.3727218 | 15.0558325 | 251.8404885 |
| **56-65** | 100.5798848 | 14.64203455 | 251.952572 |

**Table: 2.2.2** Monthly Customer Behavior Analysis by Region

| Region | Monthly Average Purchase Frequency | Average Order Value per Purchase (USD) |
|---|---|---|
| **Canada** | 14.7537092 | 249.6015628 |
| **Germany** | 14.77322677 | 254.042048 |
| **India** | 14.94995188 | 250.8801251 |
| **UK** | 14.68762475 | 249.1412575 |
| **USA** | 14.58183738 | 250.9742133 |

**Table: 2.2.2** Monthly Customer Behavior Analysis by Platform and Region

| Platform | Region | Monthly Average Purchase Frequency | Average Order Value per Purchase (USD) | Platform User Percentage (%) |
|---|---|---|---|---|
| **Facebook** | Canada | 14.56910569 | 259.414065 | 24.3 |
| **Facebook** | Germany | 14.60869565 | 252.7484585 | 25.3 |
| **Facebook** | India | 15.63485477 | 246.1695021 | 23.2 |
| **Facebook** | UK | 14.95983936 | 252.0124498 | 24.9 |
| **Facebook** | USA | 14.78629032 | 255.7689113 | 26.2 |
| **Instagram** | Canada | 15.3307393 | 250.2810117 | 25.4 |
| **Instagram** | Germany | 14.49433962 | 252.8698491 | 26.5 |
| **Instagram** | India | 14.16791045 | 255.6184701 | 25.8 |
| **Instagram** | UK | 14.84150943 | 255.2929057 | 26.4 |
| **Instagram** | USA | 14.37804878 | 242.1460569 | 26 |
| **LinkedIn** | Canada | 15.16981132 | 244.4470189 | 26.2 |
| **LinkedIn** | Germany | 15.41284404 | 247.8966055 | 21.8 |
| **LinkedIn** | India | 14.51153846 | 246.5195385 | 25 |
| **LinkedIn** | UK | 14.73858921 | 250.9095436 | 24.1 |
| **LinkedIn** | USA | 14.50226244 | 251.9811312 | 23.3 |
| **Twitter** | Canada | 13.87654321 | 244.570535 | 24 |
| **Twitter** | Germany | 14.68301887 | 261.5047547 | 26.5 |
| **Twitter** | India | 15.53703704 | 254.5806296 | 26 |
| **Twitter** | UK | 14.19838057 | 237.9215385 | 24.7 |
| **Twitter** | USA | 14.65517241 | 254.2505603 | 24.5 |

Canada
Germany
India
UK
USA

**Table: 2.2.2** Distribution of Average Order Value Ranges

| Order Value Range (USD) | Number of Orders |
|---|---|
| **5-20** | 136 |
| **21-50** | 287 |
| **51-100** | 510 |
| **101-200** | 1025 |
| **201-300** | 1051 |
| **301-400** | 1038 |
| **401-500** | 953 |

**Table: 2.2.2** Mean Average Order Value by Platform

| Platform | Mean Average Order Value (USD) |
|---|---|
| **Facebook** | 253.2496847 |
| **Instagram** | 251.3904919 |
| **LinkedIn** | 248.192556 |
| **Twitter** | 250.7708194 |

## 2.3 Campaign Performance and Marketing Effectiveness

This section analyses the performance of individual marketing campaigns and the effectiveness of different marketing channels. The focus is placed on advertising expenditures, conversion rates, and return on investment (ROI) indicators. The findings are illustrated through a series of tables and figures to facilitate better understanding.

**Table: 2.3.1** Ad Spend and ROI per Campaign (Sorted by ROI)

| Campaign ID | Ad Spend (USD) | ROI (%) |
|---|---|---|
| **CAMPAIGN_17** | 8758.45 | -48.59 |
| **CAMPAIGN_15** | 9887.28 | -46.14 |
| **CAMPAIGN_12** | 1778.46 | 4.38 |
| **CAMPAIGN_3** | 4196.83 | 7.19 |
| **CAMPAIGN_7** | 8571.34 | 16.03 |
| **CAMPAIGN_16** | 8936.96 | 18.05 |
| **CAMPAIGN_2** | 4568.49 | 37.59 |
| **CAMPAIGN_19** | 9037.56 | 37.66 |
| **CAMPAIGN_8** | 835.02 | 39.29 |
| **CAMPAIGN_11** | 1060.25 | 41.92 |
| **CAMPAIGN_13** | 8024.54 | 45.69 |
| **CAMPAIGN_4** | 8825.65 | 61.45 |
| **CAMPAIGN_1** | 2899.24 | 62.27 |
| **CAMPAIGN_9** | 7272.38 | 105.38 |
| **CAMPAIGN_14** | 9286.87 | 148.84 |
| **CAMPAIGN_5** | 7991.27 | 155.64 |
| **CAMPAIGN_18** | 3114.37 | 158.11 |
| **CAMPAIGN_20** | 3924.63 | 163.26 |
| **CAMPAIGN_6** | 1490.86 | 189.3 |
| **CAMPAIGN_10** | 1598.68 | 196.93 |

**Table: 2.3.1** Ad Spend, ROI, and Platform per Campaign (Sorted by ROI)

| Campaign ID | Ad Spend (USD) | ROI (%) | Marketing Channel |
|---|---|---|---|
| **CAMPAIGN_17** | 8758.45 | -48.59 | TV Ads |
| **CAMPAIGN_15** | 9887.28 | -46.14 | Google Ads |
| **CAMPAIGN_12** | 1778.46 | 4.38 | TV Ads |
| **CAMPAIGN_3** | 4196.83 | 7.19 | TV Ads |
| **CAMPAIGN_7** | 8571.34 | 16.03 | Google Ads |
| **CAMPAIGN_16** | 8936.96 | 18.05 | Google Ads |
| **CAMPAIGN_2** | 4568.49 | 37.59 | TV Ads |
| **CAMPAIGN_19** | 9037.56 | 37.66 | TV Ads |
| **CAMPAIGN_8** | 835.02 | 39.29 | Email |
| **CAMPAIGN_11** | 1060.25 | 41.92 | Social Media |
| **CAMPAIGN_13** | 8024.54 | 45.69 | TV Ads |
| **CAMPAIGN_4** | 8825.65 | 61.45 | TV Ads |
| **CAMPAIGN_1** | 2899.24 | 62.27 | Social Media |
| **CAMPAIGN_9** | 7272.38 | 105.38 | Google Ads |
| **CAMPAIGN_14** | 9286.87 | 148.84 | Social Media |
| **CAMPAIGN_5** | 7991.27 | 155.64 | TV Ads |
| **CAMPAIGN_18** | 3114.37 | 158.11 | Google Ads |
| **CAMPAIGN_20** | 3924.63 | 163.26 | Email |
| **CAMPAIGN_6** | 1490.86 | 189.3 | Social Media |
| **CAMPAIGN_10** | 1598.68 | 196.93 | TV Ads |

**Table: 2.3.1** Ad Spend, ROI, Marketing Channel, and Purchasers per 100 Users (Sorted by ROI)

| Campaign ID | Ad Spend (USD) | ROI (%) | Marketing Channel | Number of Purchasers per 100 Users |
|---|---|---|---|---|
| **CAMPAIGN_17** | 8758.45 | -48.59 | TV Ads | 2.59 |
| **CAMPAIGN_15** | 9887.28 | -46.14 | Google Ads | 63.46 |
| **CAMPAIGN_12** | 1778.46 | 4.38 | TV Ads | 5.51 |
| **CAMPAIGN_3** | 4196.83 | 7.19 | TV Ads | 91.4 |
| **CAMPAIGN_7** | 8571.34 | 16.03 | Google Ads | 47.56 |
| **CAMPAIGN_16** | 8936.96 | 18.05 | Google Ads | 93.23 |
| **CAMPAIGN_2** | 4568.49 | 37.59 | TV Ads | 11.38 |
| **CAMPAIGN_19** | 9037.56 | 37.66 | TV Ads | 94.36 |
| **CAMPAIGN_8** | 835.02 | 39.29 | Email | 49.16 |
| **CAMPAIGN_11** | 1060.25 | 41.92 | Social Media | 46.56 |
| **CAMPAIGN_13** | 8024.54 | 45.69 | TV Ads | 29.67 |
| **CAMPAIGN_4** | 8825.65 | 61.45 | TV Ads | 45.23 |
| **CAMPAIGN_1** | 2899.24 | 62.27 | Social Media | 84.78 |
| **CAMPAIGN_9** | 7272.38 | 105.38 | Google Ads | 94.04 |
| **CAMPAIGN_14** | 9286.87 | 148.84 | Social Media | 89.74 |
| **CAMPAIGN_5** | 7991.27 | 155.64 | TV Ads | 25.23 |
| **CAMPAIGN_18** | 3114.37 | 158.11 | Google Ads | 75.59 |
| **CAMPAIGN_20** | 3924.63 | 163.26 | Email | 44.83 |
| **CAMPAIGN_6** | 1490.86 | 189.3 | Social Media | 44.64 |
| **CAMPAIGN_10** | 1598.68 | 196.93 | TV Ads | 50.17 |

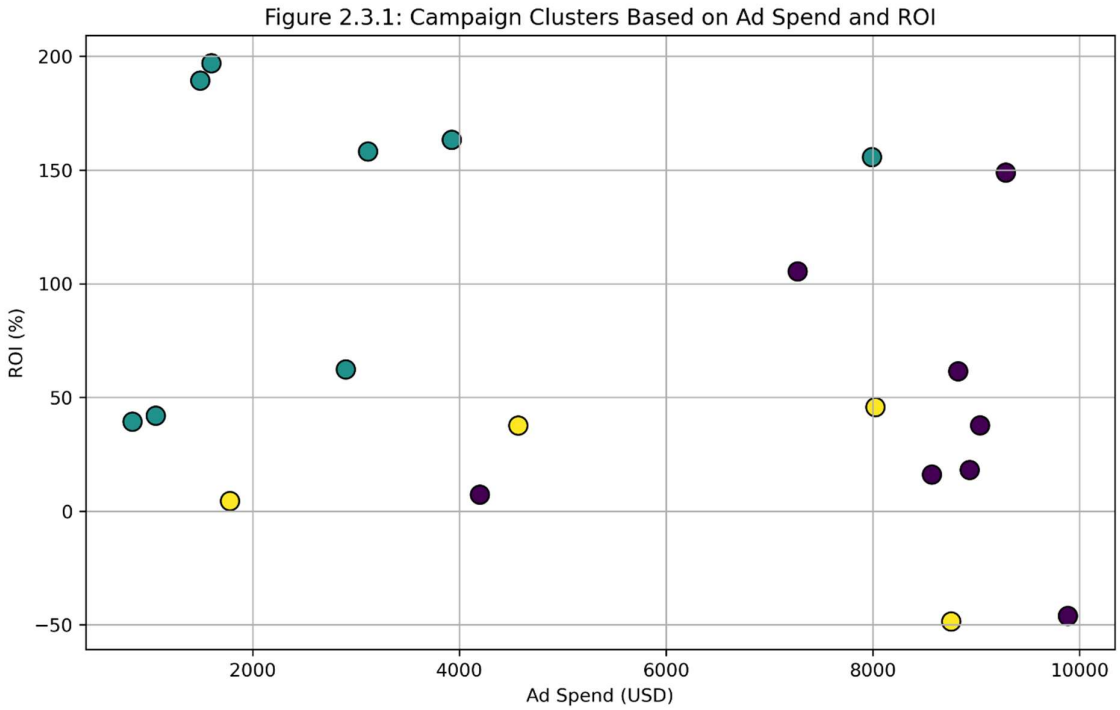


**Figure 2.3.1**: Campaign Clusters Based on Ad Spend and ROI

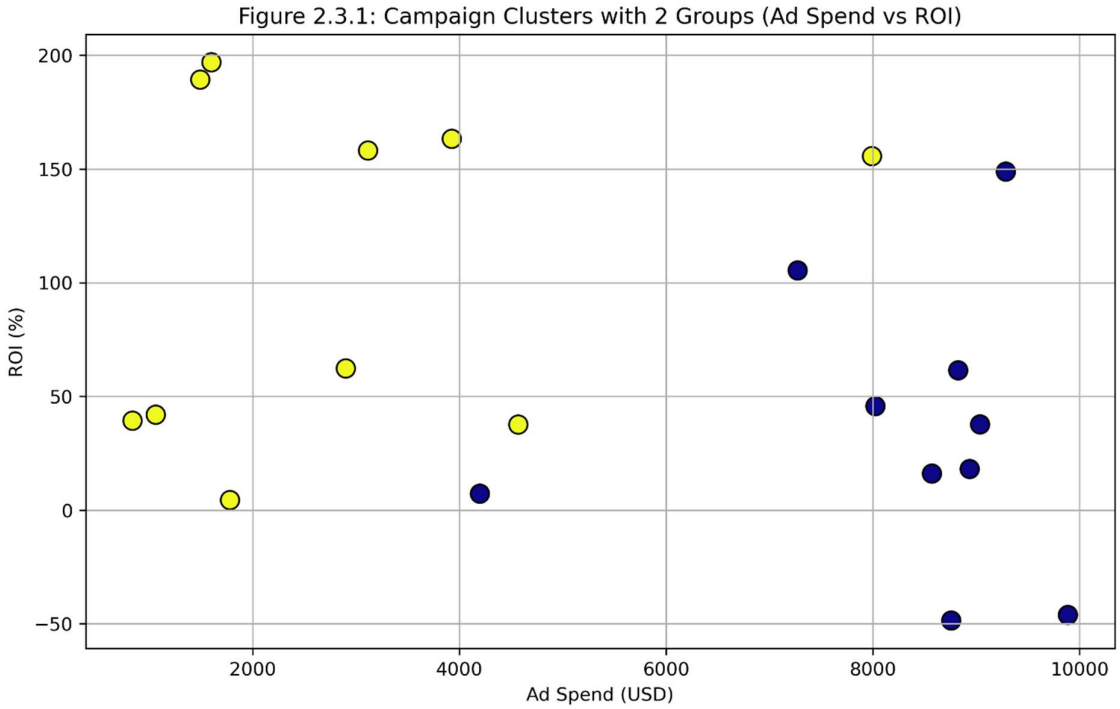


**Figure 2.3.1**: Campaign Clusters with 2 Groups (Ad Spend vs ROI)